

SINES MONTES CHAOS, PortugalWMO: **085410**Lat: **37.9544N**Lon: **8.8383W**Elev: **99**StdP: **100.14**Time Zone: **0.00 (EUW)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	6.0	-3.8	2.8	7.7	-0.9	3.5	8.6	12.1	11.8	10.7	11.7	3.8	10	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	29.3	18.8	27.4	18.6	25.8	18.4	20.7	25.1	20.2	24.1	19.7	23.4	4.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	14.5	21.5	19.0	14.0	21.1	18.5	13.5	20.7	60.2	25.3	58.1	24.3	56.4	23.4	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.2	8.2	1.7	34.3	1.6	2.8	0.5	36.4	-0.4	38.0	-1.4	39.6	-2.5	41.7		
			DB				-1.5	23.2	-2.5	24.1	-3.4	24.9	-4.6	26.1		
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	11.7	11.8	13.5	14.8	17.0	19.3	20.8	20.0	18.4	14.5	12.5
		DBStd	3.94	2.16	2.09	2.15	2.17	2.48	2.18	1.82	1.99	1.71	2.27	2.20
		HDD10.0	25	10	7	2	0	0	0	0	0	0	1	5
		HDD18.3	1048	207	182	151	109	58	12	1	3	26	118	182
		CDD10.0	2292	62	59	109	143	217	278	311	336	302	261	134
		CDD18.3	273	0	0	1	3	16	40	54	77	54	28	0
		CDH23.3	1111	0	0	6	16	93	181	221	347	162	84	1
		CDH26.7	252	0	0	1	2	22	39	58	93	23	14	0
(14)	Wind	WSAvg	4.0	3.8	4.1	4.2	4.1	4.2	4.2	4.6	4.0	3.6	3.6	3.9
(15)	Precipitation	PrecAvg	510	68	58	47	48	33	8	1	3	24	76	85
(16)		PrecMax	795	270	168	142	137	80	36	4	20	87	148	272
(17)		PrecMin	317	1	4	3	5	0	0	0	1	7	5	4
(18)		PrecStd	118	51	40	34	31	26	10	1	6	25	43	65
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	18.0	19.0	23.9	25.3	29.8	30.6	31.5	32.6	29.7	28.6	22.5	19.3
(20)		0.4% MCWB	13.5	14.7	14.7	16.7	18.1	19.6	19.6	19.3	18.9	17.4	16.8	14.7
(21)		2% DB	16.5	17.3	20.7	22.6	25.7	27.3	28.0	29.0	27.0	25.5	20.2	17.5
(22)		2% MCWB	13.9	13.4	14.3	15.9	17.2	18.8	18.8	19.1	18.4	17.6	16.5	14.8
(23)		5% DB	15.7	16.2	18.5	20.5	23.3	25.2	25.5	26.8	25.2	23.5	19.1	16.7
(24)		5% MCWB	13.6	13.3	13.9	15.5	17.0	18.3	18.9	19.1	18.5	17.7	16.0	14.7
(25)		10% DB	15.0	15.2	16.9	18.6	21.2	23.6	23.8	24.8	23.7	22.0	18.1	15.9
(26)		10% MCWB	13.2	13.1	13.6	14.9	16.5	18.2	18.6	19.1	18.5	17.4	15.5	14.2
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	15.9	15.8	16.7	18.2	19.7	21.1	21.2	21.9	21.4	20.4	19.0	16.7
(28)		0.4% MCDB	16.6	17.2	20.2	22.8	25.1	26.2	26.5	26.3	24.8	23.0	20.1	17.1
(29)		2% WB	15.0	14.9	15.5	16.9	18.6	20.0	20.2	20.7	20.3	19.7	17.9	16.0
(30)		2% MCDB	15.7	15.9	18.0	20.8	23.5	25.2	25.0	25.4	23.3	21.8	19.3	16.8
(31)		5% WB	14.3	14.3	14.8	16.1	17.7	19.2	19.5	20.1	19.7	19.0	17.0	15.2
(32)		5% MCDB	15.2	15.5	16.9	19.3	21.9	23.8	23.8	24.4	22.5	21.2	18.2	16.3
(33)		10% WB	13.6	13.5	14.2	15.2	16.9	18.5	18.9	19.5	19.2	18.4	16.2	14.4
(34)		10% MCDB	14.7	14.9	16.2	17.9	20.2	22.6	22.9	23.5	22.1	20.6	17.6	15.7
(35)	Mean Daily Temperature Range	MDBR	5.9	6.5	6.9	7.0	7.7	7.9	7.9	8.3	7.4	6.9	6.3	6.1
(36)		5% DB MCDBR	6.1	7.4	9.4	10.3	11.6	11.0	10.9	11.4	9.8	9.3	6.8	6.3
(37)		5% WB MCWBR	4.5	5.1	5.7	5.8	5.4	4.5	4.4	4.3	4.1	4.6	4.5	4.6
(38)		5% WB MCDBR	4.7	6.0	7.1	8.5	9.4	9.3	8.9	8.8	7.2	6.6	5.4	5.1
(39)		5% WB MCWBR	4.1	4.6	5.0	5.3	4.9	4.3	4.2	4.0	3.6	3.9	4.3	4.4
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.343	0.374	0.374	0.382	0.386	0.371	0.386	0.383	0.364	0.340	0.330
(41)		taud	2.508	2.467	2.385	2.396	2.400	2.405	2.435	2.404	2.429	2.494	2.520	2.508
(42)		Ebn at Noon	854	886	887	905	900	892	903	882	866	850	833	824
(43)		Edh at Noon	79	93	111	116	118	117	113	114	105	90	78	75
(44)	All-Sky Solar Radiation	RadAvg	2.39	3.36	4.55	5.77	6.89	7.53	7.71	6.94	5.45	3.79	2.65	2.16
(45)		RadStd	0.29	0.39	0.44	0.42	0.46	0.40	0.26	0.23	0.33	0.30	0.30	0.23

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page